

Summer Engineering Roadmap

A month of disciplined building, placement prep, and founder groundwork.

PILLAR 1	PILLAR 2	PILLAR 3
Become a Hireable Engineer	Placement Machine	Founder Education
55–60% of time — flagship AI SaaS, backend engineering, DevOps, Python	30% of time — DSA, CS core, resume, content, HR networking	15% of time — YC patterns, sales, business ops, branding, open source

What this document is

This is a working reference for the summer roadmap: the three pillars, the weekly build plan for the flagship project, the daily time budget, and the success criteria for the end of the break. The final pages are interactive — fillable checkboxes to track daily execution and milestones directly inside this PDF (open in Adobe Acrobat, Preview, or any PDF reader that supports forms).

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Daily Time Allocation (6–7 hours)

This is the daily budget the three pillars are built from. It should be treated as the default day shape, not a rigid law — but every hour should map back to one of the pillars.

Activity	Time	Pillar
DSA	1.5 hours	Pillar 2
Flagship project	2.5 hours	Pillar 1
Backend / DevOps learning tied to the project	1 hour	Pillar 1
CS Core	30 minutes	Pillar 2
LinkedIn / X or networking	20 minutes	Pillar 2
Founder research (YC, sales, workflows)	40 minutes	Pillar 3

Success Criteria by the End of Summer

A checkable version of this list is in the interactive tracker at the end of this document. Here is the plain reference version:

- **DSA:** 80–100 quality problems completed with strong understanding of core patterns.
- **Project:** Authentication, RBAC, organization management, and a clean backend architecture implemented for the AI SaaS.
- **Deployment:** One service containerized with Docker and deployed using a basic CI/CD pipeline.
- **Cloud:** Comfortable with AWS EC2, S3, IAM, and deployment basics.
- **AI:** Understand RAG, embeddings, prompt engineering, and how to integrate GenAI into workflows.
- **Professional presence:** A polished resume, active GitHub, optimized LinkedIn profile, and regular engineering posts.
- **Networking:** A repeatable outreach process with a growing list of recruiters and engineers.
- **Founder knowledge:** Notes on YC trends, enterprise workflows, and sales fundamentals that can directly influence product decisions.

The final measure: this summer should not be measured by hours studied — it should be measured by assets created: a production-grade codebase, a GitHub profile that demonstrates engineering maturity, a resume with a standout project, a disciplined DSA foundation, a growing professional network, and a personal knowledge base on AI, enterprise software, and startups.

PILLAR 1**Become a Hireable Engineer***Highest priority · ~55–60% of total time*

The objective isn't to learn more technologies — it's to become someone who can build production software. The flagship AI SaaS is the primary vehicle: treat it like you're the CTO of a startup, not like a side project.

A. Flagship AI SaaS — 4-week build arc

Week	Focus
Week 1	Backend architecture · Authentication · Organizations · RBAC · Folder architecture
Week 2	Document ingestion · Storage · Search · Queueing
Week 3	AI workflows · RAG · Embeddings · Prompt engineering
Week 4	Docker · CI/CD · Deployment · Monitoring · Production polish

This single project should teach: Express, Node, MongoDB, PostgreSQL (later), Redis, Docker, AWS basics, GitHub Actions, testing, logging, clean architecture, and AI integration.

B. Backend Engineering — master:

- Middleware
- Authentication
- Authorization
- Rate limiting
- Validation
- Background jobs
- Queues
- Caching
- API versioning

C. DevOps — enough to confidently discuss in interviews:

- Docker
- GitHub Actions
- CI/CD
- Environment management
- Nginx basics
- AWS EC2
- S3
- IAM
- Cloud deployment

D. Python — only what actually helps:

- Automation
- AI scripts
- FastAPI basics
- Data processing
- SDK usage

Don't spend weeks on Python syntax.

PILLAR 2**Placement Machine***~30% of total time · separate from project work***DSA — daily, 3–4 quality problems/day**

Understand patterns, not memorize solutions. Aim for 150–200 well-understood problems, not 500 rushed ones.

Progression: Arrays → Hashing → Strings → Sliding Window → Binary Search → Linked Lists → Trees → Graphs → Heap → DP

CS Core — rotate daily

- OS
- DBMS
- CN
- OOP

Build concise revision notes as you go.

Resume — by month end

- One-page polished resume
- Flagship project featured prominently
- Quantified impact
- GitHub links, LinkedIn

LinkedIn — twice a week

Post about engineering lessons, architecture diagrams, AI experiments, and things you've built. Avoid generic motivational content.

X — use it as an engineering notebook

- Bugs solved
- Interesting repositories
- System design notes
- AI findings

Consistency matters more than going viral.

HR Networking — once your resume is ready, every week:

- Reach out to 15–20 recruiters or engineers
- Personalize every message
- Keep a spreadsheet of outreach and responses

PILLAR 3**Founder Education**

~15% of total time · compounds over years · 45–60 min/day

YC — study patterns, not isolated ideas

- B2B SaaS
- Vertical AI
- Enterprise workflows

Look for recurring patterns across companies, not one-off ideas.

Sales — understand (engineers often neglect this)

- ICP (Ideal Customer Profile)
- Outbound
- Inbound
- Discovery calls
- Objection handling

Business Operations — research (informs enterprise product ideas)

- Procurement
- HR
- CRM
- ERP
- Finance workflows

Branding — observe

- How companies position themselves
- Pricing strategies
- Landing pages
- Onboarding flows

Open Source — don't try to become a major contributor immediately

- Read high-quality repositories
- Fix documentation
- Solve beginner issues
- Learn engineering standards

Learning Order

Don't learn technologies randomly — learn them in the order the flagship project needs them:

1. Backend architecture	2. Authentication	3. MongoDB
4. Docker	5. GitHub Actions	6. AWS
7. AI workflows	8. RAG	9. Redis
10. Queues	11. Monitoring	12. Open source

Skills That Give You Maximum ROI

If you become genuinely good at these by the end of the year, you'll be well positioned for both placements and startups:

- Clean backend architecture
- DSA
- React ecosystem
- Docker
- AWS fundamentals
- GitHub Actions
- RAG systems
- AI orchestration
- Authentication and security
- System design
- Communication and documentation

The next two pages are interactive: fillable checkboxes for daily execution and for tracking the end-of-summer success criteria. Open this PDF in a form-capable reader (Adobe Acrobat/Reader, Preview on macOS, or most modern browsers) to check items off directly.

Interactive Daily Execution Checklist

Check off today's blocks as you complete them. Reusable — uncheck at the start of a new day.

Date:

Day # of month:

PILLAR 1 · HIREABLE ENGINEER (4 h)

Flagship project — 2.5h build block (this week's focus area)

Backend / DevOps learning tied directly to today's project work — 1h

PILLAR 2 · PLACEMENT MACHINE (2h 20m)

DSA — 3–4 quality problems, 1.5h (log pattern, not just solved/unsolved)

CS Core rotation — 30 min (OS / DBMS / CN / OOP)

LinkedIn or X post, or HR outreach batch — 20 min

PILLAR 3 · FOUNDER EDUCATION (40 min)

YC / vertical AI / enterprise workflow reading

Sales fundamentals or business-ops research

Branding / positioning observation

END-OF-DAY REVIEW

Committed code / pushed to GitHub

Logged today's DSA problems and pattern notes

Noted one thing that blocked me for 20+ min (and how I resolved it)

Tomorrow's top project task is written down

One-line note for today:

Interactive Weekly Tracker

One row per week of the break. Track flagship-project milestones and DSA volume together.

WEEK	FLAGSHIP PROJECT FOCUS	DSA DONE	MILESTONES HIT
Week 1	Backend architecture · Auth · Organizations · RBAC · Folder architecture	problems	On track Deployed/tested Documented
Week 2	Document ingestion · Storage · Search · Queueing	problems	On track Deployed/tested Documented
Week 3	AI workflows · RAG · Embeddings · Prompt engineering	problems	On track Deployed/tested Documented
Week 4	Docker · CI/CD · Deployment · Monitoring · Production polish	problems	On track Deployed/tested Documented

Weekly reflection

Interactive Success Criteria Tracker

Check each box as the criterion is genuinely met — not just attempted.

DSA

80–100 quality problems completed with strong understanding of core patterns.

Project — Auth & RBAC

Authentication, RBAC, and organization management implemented.

Project — Architecture

Clean backend architecture in place across the flagship SaaS.

Deployment

One service containerized with Docker and deployed via a basic CI/CD pipeline.

Cloud

Comfortable with AWS EC2, S3, IAM, and deployment basics.

AI

Understand RAG, embeddings, prompt engineering, and GenAI workflow integration.

Resume

Polished one-page resume with the flagship project featured prominently.

GitHub

Active GitHub profile that demonstrates engineering maturity.

LinkedIn

Optimized profile with regular engineering posts (2x/week cadence hit).

Networking

A repeatable outreach process with a growing recruiter/engineer list.

Founder notes

Written notes on YC trends, enterprise workflows, and sales fundamentals.

Open source

At least one documentation fix or beginner issue solved in a real repo.
